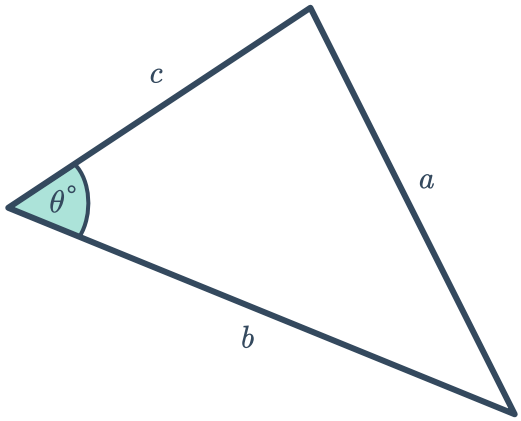
# Cosine rule

**1** Write an expression for $\cos\theta$ using the cosine rule for the following triangle:

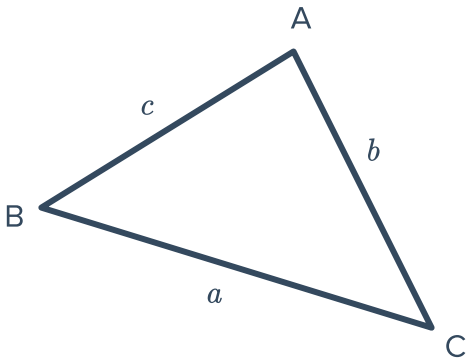



**2** Consider the triangle given below:

**a** Write an expression for $\cos A$ using the cosine rule.

**b** Write an expression for $a^2$ using the cosine rule.

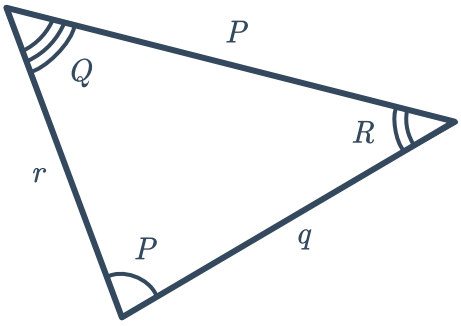
**c** Write an expression for $b$ using the cosine rule.




**3** Consider the triangle given below:

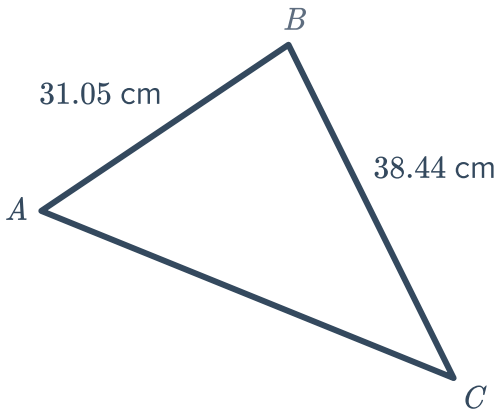
**a** Write an expression for $\cos Q$ using the cosine rule.

**b** Write an expression for $r^2$ using the cosine rule.




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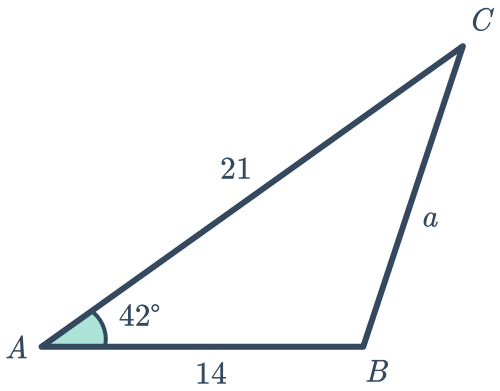
**4** To use the cosine rule to find the length of $AC$, which angle would need to be given?



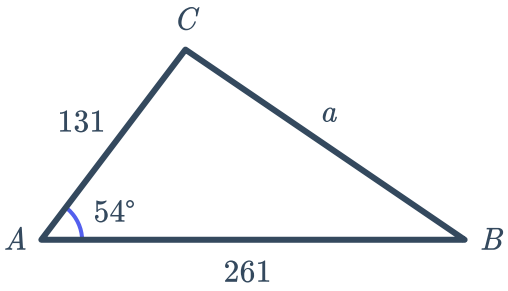

## Unknown sides

**5** Find the length of the missing side in each of the following triangles using the cosine rule. Round your answers to two decimal places.

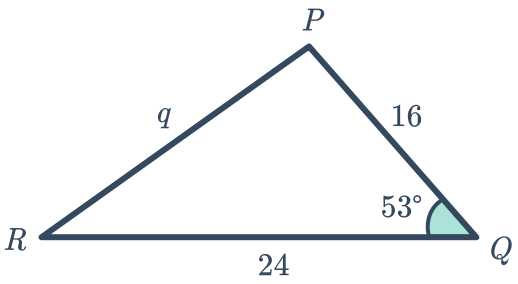
**a**



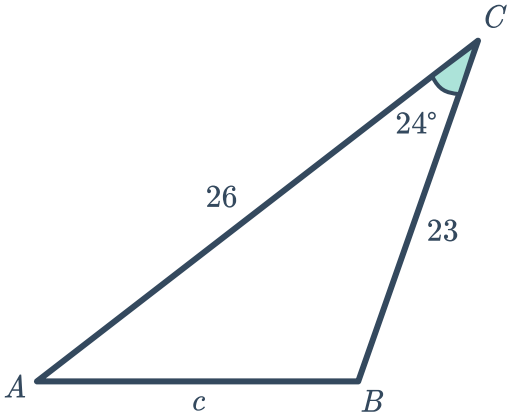

**b**




**c**

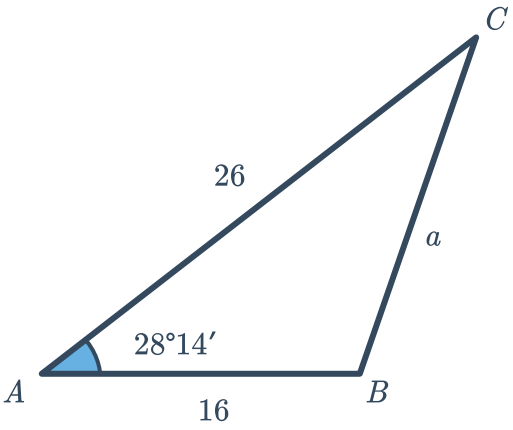



**d**

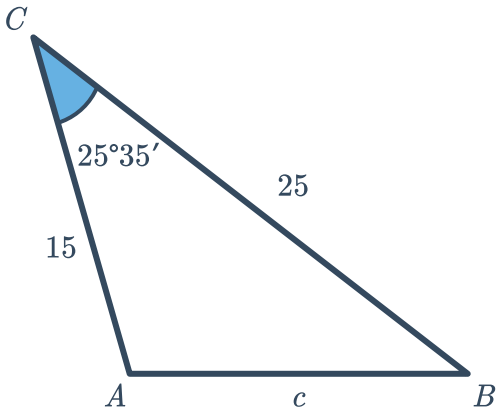



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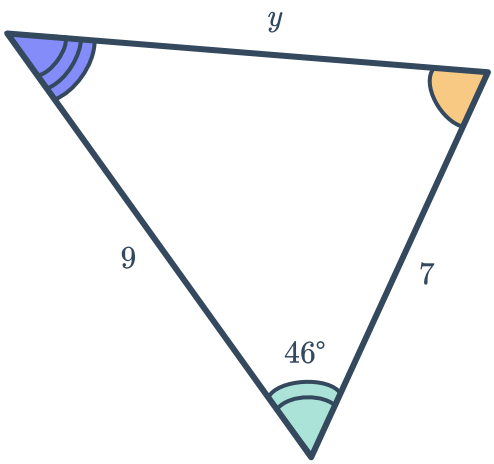
e



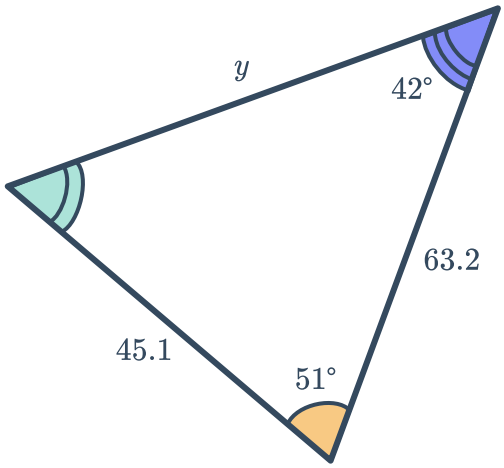

f



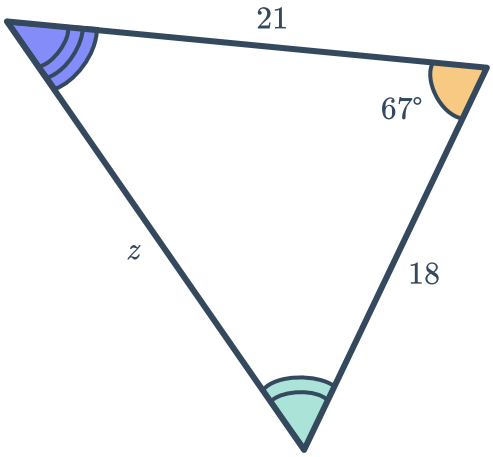

g



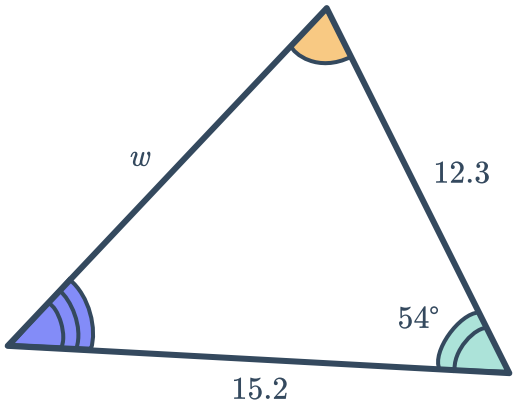

h




i

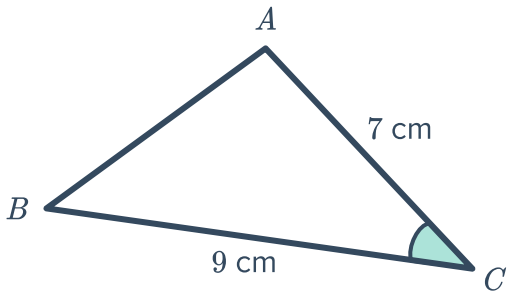



j




6 In $\triangle ABC$, $\cos C = \frac{8}{9}$:
Find the exact length of side $AB$ in centimetres.

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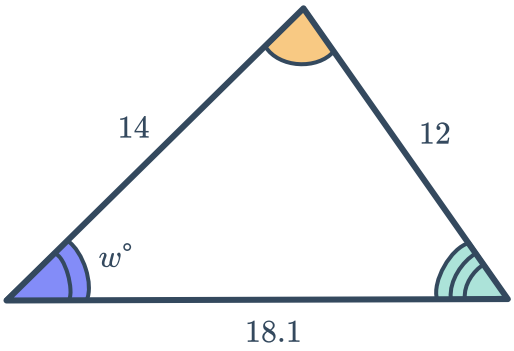

**7** In $\triangle QUV$, $q = 5$, $u = 6$ and $\cos V = \frac{3}{5}$. Find the value of $v$.

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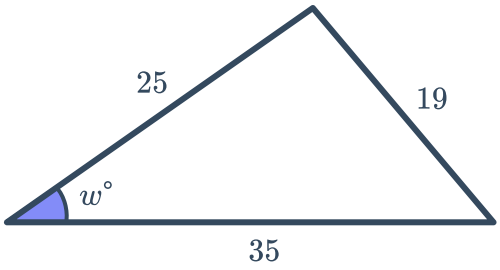
## Unknown angles

**8** For each of the following triangles, find the value of the pronumeral to the nearest degree:

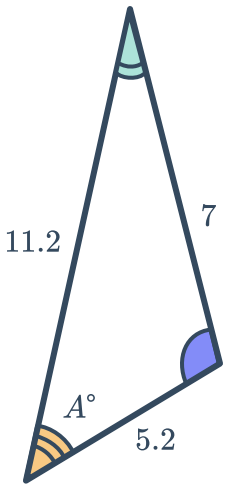
**a**



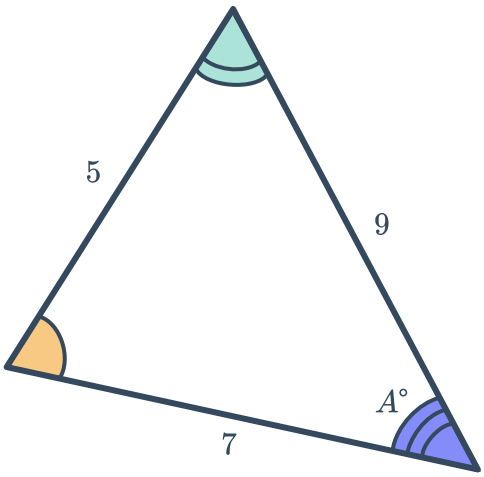

**b**




**c**

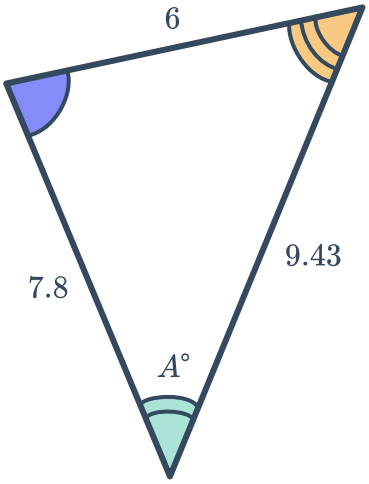



**d**

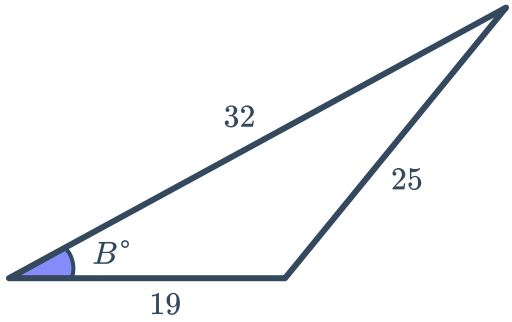



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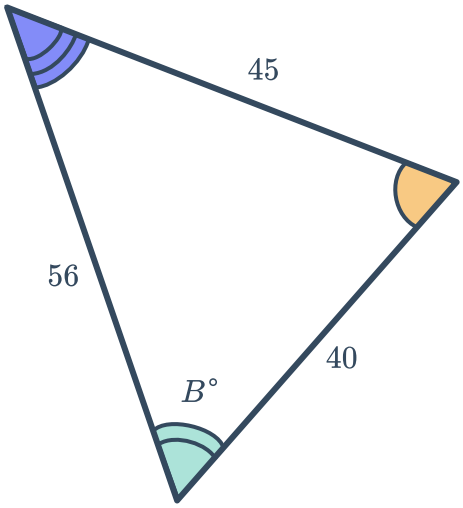
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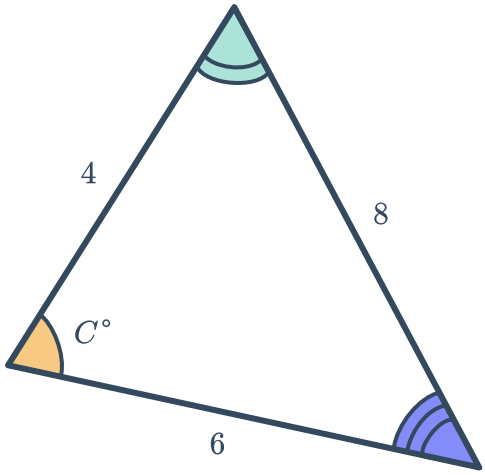

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g

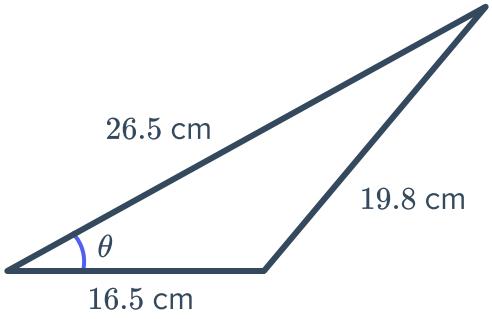



h

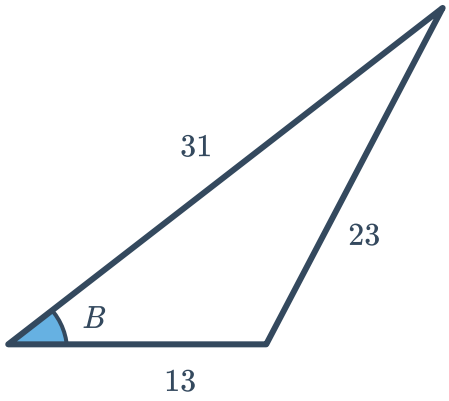



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**9** Find the value of $\theta$ in the following triangle. Round your answer to the nearest hundredth of a degree.




**10** Find the value of $B$ in the following triangle. Round your answer to the nearest second.




**11** In $\triangle QUV$, $v = 8$, $u = 9$ and $q = 15$. Solve for $\cos Q$.

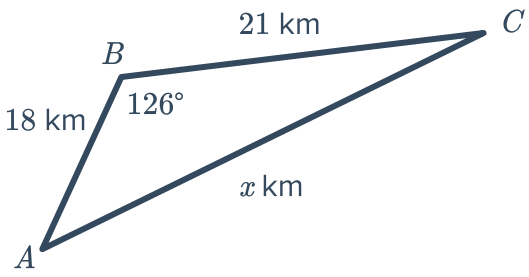
**12** The sides of a triangle are in the ratio $4 : 5 : 8$. Find $\theta$, the smallest angle in the triangle, to the nearest degree.

**13** A triangle has sides of length $11\ \text{cm}$, $18\ \text{cm}$ and $8\ \text{cm}$. Find $x$, the largest angle in the triangle, to the nearest degree.

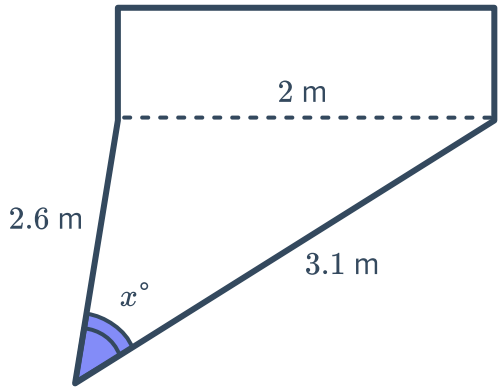
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## Applications

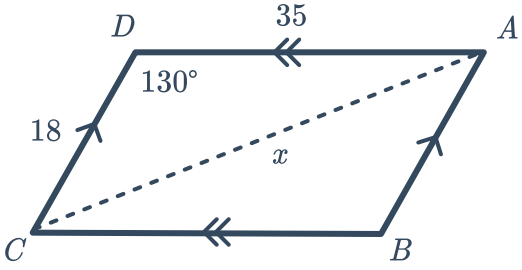
**14** Mae went for a bike ride on Sunday morning from Point $A$ to Point $B$, which was $18\ \text{km}$ long. She then took a $126^\circ$ turn and rode from Point $B$ to Point $C$, which was $21\ \text{km}$ long.
Find $x$, the distance in kilometres from her starting point to Point $C$ to two decimal places.




**15** A goal has posts that are $2\ \text{m}$ apart. Buzz shoots for the goal when he is $2.6\ \text{m}$ from one post and $3.1\ \text{m}$ from the other post.
Find the size of the angle, $x$, in which he can score a goal. Round your answer to the nearest degree.

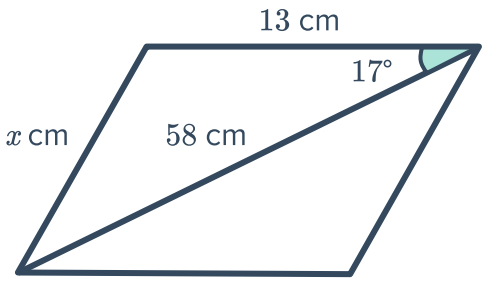



**16** Find the length of the diagonal, $x$, in parallelogram $ABCD$.
Round your answer to two decimal places.




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**17** Consider the parallelogram in the given diagram that has a side of length $13\ \text{cm}$ and a diagonal of length $58\ \text{cm}$:
Find the value of $x$. Round your answer to one decimal place.

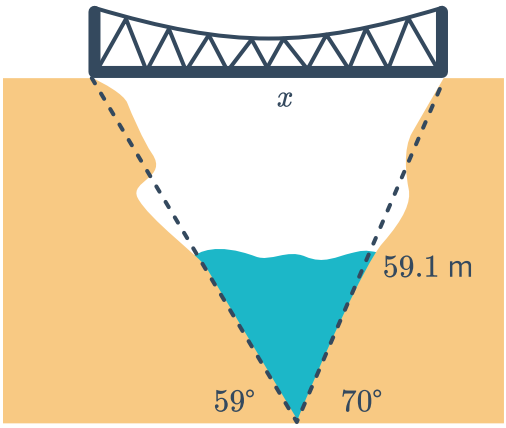



**18** In a sailing boat race, teams must start at buoy $A$ and navigate around buoys $B$ and $C$ before returning to buoy $A$ to cross the line. The first leg of the race is $100.6\ \text{km}$ long, the second leg of the race is $190.3\ \text{km}$ long, and the angle between these legs is $143^\circ$.

**a** Find $x$, the distance of the third leg of the race, correct to one decimal place.

**b** Hence, find the total length of the race, correct to one decimal place.

**19** A bridge connects two towns on either side of a gorge, where one side of the gorge is inclined at $59^\circ$ and the other side is inclined at $70^\circ$. The length of the steeper incline is $59.1\ \text{m}$.
Find $x$, the length of the bridge. Round your answer correct to one decimal place.




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